are designed to be implemented securely against side-channel attacks. Not all JWT libraries or AS software supports EdDSA signatures yet. The older ECDSA standard for elliptic curve digital signatures has wider support, and shares some of the same properties as EdDSA, but is slightly slower and harder to implement securely.

> **WARNING** ECDSA signatures require a unique random nonce for each signature. If a nonce is repeated, or even just a few bits are not completely random, then the private key can be reconstructed from the signature values. This kind of bug was used to hack the Sony Playstation 3, steal Bitcoin cryptocurrency from wallets on Android mobile phones, among many other cases. Deterministic ECDSA signatures (https://tools.ietf.org/html/rfc6979) can be used to prevent this, if your library supports them. EdDSA signatures are also immune to this issue.

RSA signatures are expensive to produce, especially for secure key sizes (a 3072-bit RSA key is roughly equivalent to a 256-bit elliptic curve key or a 128-bit HMAC key) and produce much larger signatures than the other options, resulting in larger JWTs. The variants of RSA using PSS padding should be preferred over those using the older PKCS#1 version 1.5 padding but may not be supported by all libraries.

7.4.5 Encrypted JWT access tokens